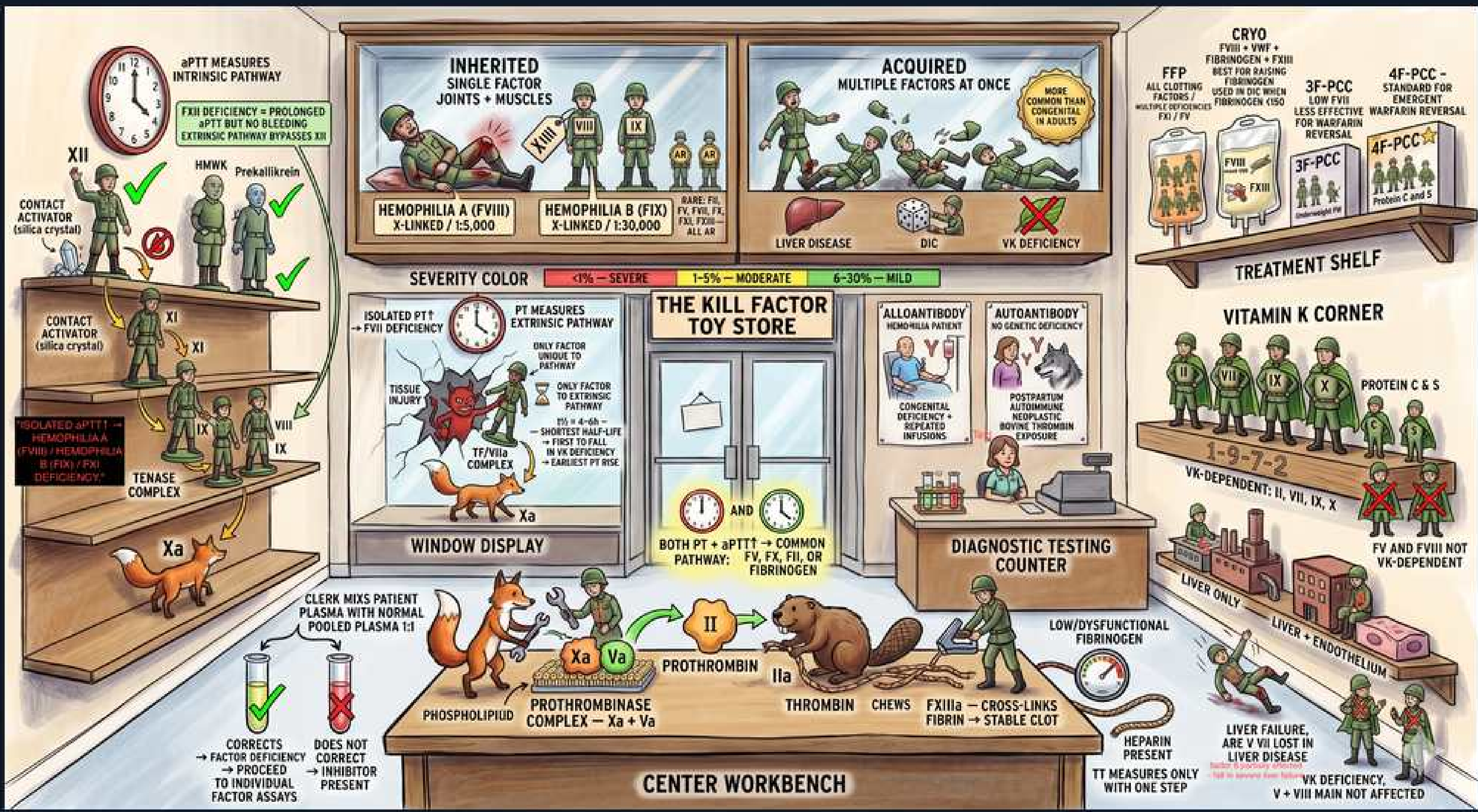
Coagulation — Cascade Overview



1. Intrinsic pathway / aPTT

WHAT YOU SEE

A wall clock at top-left labelled aPTT — the clock = the timed test, here measuring the intrinsic pathway

WHAT IT ENCODES

aPTT (activated partial thromboplastin time) measures the INTRINSIC pathway — factors XII, XI, IX, VIII — plus the common pathway. A prolonged aPTT with normal PT localizes the defect to an intrinsic-only factor (XII, XI, IX, or VIII). aPTT is also the test used to monitor unfractionated heparin.

BOARD TRAP

aPTT does NOT measure the extrinsic factor VII. An isolated long aPTT cannot be from FVII deficiency — that gives an isolated long PT instead.

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2. Factor XII contact activation

WHAT YOU SEE

A green soldier figure 'XII contact activator' on the left shelf beside prekallikrein/HMWK — contact-activation kicks off the intrinsic limb

WHAT IT ENCODES

The intrinsic cascade begins with the contact-activation complex: factor XII, high-molecular-weight kininogen (HMWK), and prekallikrein, triggered on a negatively charged surface. These activate XI → IX. Deficiencies markedly prolong aPTT.

BOARD TRAP

Factor XII (and HMWK/prekallikrein) deficiency prolongs the aPTT but causes NO clinical bleeding — a long aPTT in an asymptomatic patient with no bleeding history points here, not to hemophilia.

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3. Tenase complex (VIIIa-IXa)

WHAT YOU SEE

'Tenase complex' shelf with Xa being generated on the left

WHAT IT ENCODES

The intrinsic tenase complex = factor IXa + cofactor VIIIa assembled on phospholipid with Ca^{2}□, which activates factor X to Xa. This is the amplification step hit by hemophilia A (VIII deficiency) and hemophilia B (IX deficiency), both X-linked recessive.

BOARD TRAP

Tenase is part of the INTRINSIC pathway (aPTT), not the extrinsic pathway. Hemophilia A and B therefore prolong aPTT with a NORMAL PT.

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4. Extrinsic pathway / PT-INR

WHAT YOU SEE

A fox in the 'extrinsic pathway' window display holding factor VII — the fox/window scene = the PT/INR extrinsic limb

WHAT IT ENCODES

PT/INR measures the EXTRINSIC pathway: tissue factor (TF, released from injured tissue) + factor VII → activate X. PT is most sensitive to vitamin K status and is the test used to monitor warfarin (reported as INR to standardize across labs).

BOARD TRAP

PT does NOT measure the intrinsic factors XII/XI/IX/VIII. A normal PT does not rule out hemophilia (those are detected by aPTT).

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5. Isolated PT = factor VII

WHAT YOU SEE

A note reading 'isolated PT prolongation = FVII deficiency' — flags the one factor unique to the extrinsic pathway

WHAT IT ENCODES

An isolated prolonged PT with a normal aPTT points to factor VII — the only factor unique to the extrinsic pathway. FVII has the SHORTEST half-life (~4-6 h) of all clotting factors, so it falls first in early vitamin K deficiency, warfarin initiation, or liver disease → PT/INR is the earliest test to become abnormal.

BOARD TRAP

Isolated long PT is from FVII (or early vit-K/warfarin effect), NOT from factor VIII (which would prolong aPTT).

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6. Common pathway (X,V,II,I)

WHAT YOU SEE

Center workbench: prothrombinase Xa+Va, prothrombin II to thrombin, fibrinogen to fibrin

WHAT IT ENCODES

The common pathway = factor X, V, II (prothrombin), and I (fibrinogen). Prothrombinase (Xa + cofactor Va on phospholipid) converts prothrombin (II) → thrombin (IIa), which cleaves fibrinogen (I) → fibrin. Both PT and aPTT feed into this final shared limb.

BOARD TRAP

A defect prolonging BOTH PT and aPTT is in the common pathway (or reflects vit-K deficiency, liver disease, DIC, or multiple-factor problems) — not an isolated intrinsic or extrinsic lesion.

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7. Both PT + aPTT = common

WHAT YOU SEE

A central sign 'BOTH PT + aPTT = COMMON pathway (X, V, II, I, fibrinogen)' — both clocks point to the shared final limb

WHAT IT ENCODES

When BOTH PT and aPTT are prolonged, the lesion is in the common pathway (X, V, II, fibrinogen) OR there is a global problem: vitamin K deficiency/warfarin (II,VII,IX,X), liver disease, DIC, heparin (high dose), or a direct thrombin/Xa inhibitor.

BOARD TRAP

'Both prolonged' is NOT specific for hemophilia (which prolongs aPTT only). Use the pattern: long aPTT only = intrinsic; long PT only = extrinsic; both = common/global.

8. Thrombin → fibrin

WHAT YOU SEE

Thrombin (IIa) crocodile chewing fibrinogen into fibrin at bottom center

WHAT IT ENCODES

Thrombin (IIa) is the cascade's central enzyme: it cleaves fibrinogen → fibrin monomers, activates cofactors V and VIII and platelets (positive feedback), and activates factor XIII. Factor XIIIa then cross-links fibrin into a stable, insoluble clot.

BOARD TRAP

Factor XIII deficiency gives a NORMAL PT and aPTT (those tests stop at the first fibrin strands). It presents as delayed/recurrent bleeding, poor wound healing, and umbilical-stump bleeding; diagnose with a urea/factor XIII clot-solubility assay.

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9. Mixing study — corrects

WHAT YOU SEE

Bottom-left tubes: patient plasma mixed with normal pooled plasma, 'CORRECTS = factor deficiency'

WHAT IT ENCODES

A 1:1 mixing study with normal pooled plasma that CORRECTS the prolonged time = FACTOR DEFICIENCY (the normal plasma supplies the missing factor, only ~50% activity needed). Next step: replace/measure the specific factor.

BOARD TRAP

Correction = deficiency, NOT an inhibitor. Note an immediate correction that LATER prolongs on incubation (1-2 h at 37°C) suggests a time/temperature-dependent inhibitor such as an acquired factor VIII autoantibody.

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10. Mixing study — no correction

WHAT YOU SEE

'Does NOT correct = inhibitor present' tube

WHAT IT ENCODES

A mixing study that does NOT correct = an INHIBITOR is present, blocking even the added normal factors. Examples: acquired factor VIII antibody (acquired hemophilia — isolated long aPTT, bleeding) and lupus anticoagulant (long aPTT, paradoxical thrombosis, no bleeding).

BOARD TRAP

Failure to correct = inhibitor, not a simple deficiency. Distinguish a bleeding inhibitor (anti-FVIII) from a thrombotic one (lupus anticoagulant) by clinical picture and confirmatory phospholipid-dependent testing (dRVVT).

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11. Vitamin K factors II,VII,IX,X

WHAT YOU SEE

Right 'Vitamin K corner' soldiers labeled VK-dependent II VII IX X

WHAT IT ENCODES

Vitamin K-dependent factors are II, VII, IX, X PLUS anticoagulant proteins C and S. Vitamin K enables γ-carboxylation (via the enzyme epoxide reductase, VKORC1) so these factors can bind Ca^{2+}/phospholipid. Warfarin inhibits VKORC1; this is the basis of warfarin's effect and reversal with vitamin K.

BOARD TRAP

Factors V and VIII are NOT vitamin K-dependent. Also: because proteins C and S (anticoagulants, short half-lives) drop first, warfarin causes a transient HYPERcoagulable state at initiation — hence skin necrosis risk and the need for heparin bridging.

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12. Factor VIII not VK-dependent

WHAT YOU SEE

Note 'FV and FVIII not VK-dependent'

WHAT IT ENCODES

Factors V and VIII are not vitamin K-dependent. Factor VIII is unique: it is synthesized largely by endothelium/sinusoids (not hepatocytes) and is an ACUTE-PHASE reactant, so it rises with inflammation and stays normal or HIGH in liver disease, while all other factors fall.

BOARD TRAP

Don't list VIII among warfarin-sensitive factors. In liver disease use factor VIII to distinguish the cause of coagulopathy: VIII is preserved/elevated in liver disease but LOW in DIC (where everything is consumed) — a key discriminator.

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13. Treatment shelf (FFP/PCC/cryo)

WHAT YOU SEE

Top-right treatment shelf with FFP, cryo, 3F-PCC, 4F-PCC

WHAT IT ENCODES

Replacement options: FFP contains ALL clotting factors; cryoprecipitate is concentrated in fibrinogen, factor VIII, vWF, and factor XIII; 4-factor PCC (II, VII, IX, X + proteins C/S) gives rapid, low-volume warfarin/vitamin-K-antagonist reversal (3F-PCC lacks meaningful VII).

BOARD TRAP

For low fibrinogen (e.g., DIC), CRYOPRECIPITATE is the concentrated source — not FFP first-line. For urgent warfarin reversal with major bleeding, use 4F-PCC + IV vitamin K, NOT FFP (faster, smaller volume).

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14. Heparin → TT prolonged

WHAT YOU SEE

Bottom-right 'heparin present, TT prolonged, fibrin not cross-linked' note

WHAT IT ENCODES

Thrombin time (TT) measures the final fibrinogen → fibrin step directly. It is prolonged by heparin (potentiates antithrombin against thrombin), by low or dysfunctional fibrinogen, by fibrin degradation products (DIC), and by direct thrombin inhibitors. A normal TT excludes a significant heparin effect and clinically relevant fibrinogen problems.

BOARD TRAP

A prolonged TT from heparin is a drug effect, NOT a true clotting-factor deficiency. The reptilase time is unaffected by heparin (uses a snake-venom enzyme), so heparin contamination = TT prolonged but reptilase time normal.

15. Hemophilia A & B inherited

WHAT YOU SEE

Top-center 'INHERITED single factor — joints/muscles' panel: Hemophilia A (FVIII) X-linked 1/5,000, Hemophilia B (FIX) X-linked 1/30,000

WHAT IT ENCODES

The classic inherited single-factor disorders are X-linked recessive (affect males): Hemophilia A (factor VIII, ~1/5,000 males) and Hemophilia B / Christmas disease (factor IX, ~1/30,000). Both prolong aPTT with normal PT and present with deep bleeding into JOINTS (hemarthroses) and muscles. Severity tracks factor level: <1% severe, 1-5% moderate, 5-40% mild.

BOARD TRAP

Hemophilia (A or B) gives an ISOLATED long aPTT (normal PT, normal platelets/bleeding time) and is clinically indistinguishable between A and B without a specific factor assay. Contrast with von Willebrand disease, where mucocutaneous bleeding and a prolonged bleeding time/PFA dominate.

16. Acquired multiple factors

WHAT YOU SEE

Top-center-right 'ACQUIRED — multiple factors at once' panel: liver disease, vitamin K deficiency

WHAT IT ENCODES

Acquired coagulopathies hit MULTIPLE factors simultaneously: liver disease (reduced synthesis of all factors except VIII, plus low fibrinogen and thrombocytopenia) and vitamin K deficiency/warfarin (II, VII, IX, X + proteins C/S). Both typically prolong PT first (FVII shortest half-life), then aPTT as more factors fall.

BOARD TRAP

Use factor VIII to separate the two big consumptive/synthetic pictures: VIII is NORMAL/HIGH in liver disease (endothelial source, acute-phase) but LOW in DIC. Vitamin K deficiency corrects with vitamin K; liver-disease coagulopathy does not.

17. von Willebrand factor / VIII carrier

WHAT YOU SEE

vWF drawn cradling/carrying factor VIII at the central workbench — vWF is the chaperone that stabilizes circulating FVIII

WHAT IT ENCODES

Von Willebrand factor (vWF) carries and stabilizes circulating factor VIII (protecting it from degradation) and mediates platelet adhesion to subendothelium. Von Willebrand disease is the most common inherited bleeding disorder (autosomal, both sexes) and causes mucocutaneous bleeding; severe vWD lowers VIII secondarily, mildly prolonging aPTT.

BOARD TRAP

vWD presents with mucocutaneous/platelet-type bleeding and a long bleeding time/abnormal PFA-100 — not the deep joint bleeds of hemophilia. Desmopressin (DDAVP) releases endothelial vWF/VIII stores and treats mild type 1 vWD and mild hemophilia A, but is ineffective/contraindicated in type 2B (worsens thrombocytopenia) and useless in hemophilia B.